Sales Data Cleaning Report

Synthetic e-commerce sales dataset | Portfolio practice project | Python & Pandas

Original records	Exact duplicates removed	Final records
1,000	50	950

Executive summary

The supplied cleaned file was compared against the source data. Cleaning removed 50 exact duplicate records, standardized categories and dates, and filled 95 missing numerical cells using documented median-based estimates. Fifteen negative unit prices remain flagged for review, rather than silently overwritten.

Verified transformation summary

Operation	Verified result
Exact duplicate rows	50 removed; 0 remain
Unit_Price missing cells	55 filled with product-specific median from positive prices
Quantity missing cells	40 filled with overall median of 3
City labels	65 standardized (case and surrounding spaces)
Status labels	35 standardized (case and surrounding spaces)
Order_Date	42 standardized to YYYY-MM-DD
Negative unit prices	15 records at -5 retained and flagged
Final data quality	950 rows, 8 columns; 0 missing values; 0 exact duplicates

Date normalization

Original value	Occurrences	Converted date
2026/04/20	21	2026-04-20
15-05-2026	11	2026-05-15
03/15/2026	10	2026-03-15

Quality validation and cautions

Duplicate handling: Only exact duplicate rows were removed; 950 distinct Order_ID values remain. **Quantity:** All final values are positive whole-number quantities. **Dates:** All final date strings parse as YYYY-MM-DD. **Categories:** Five city names and three order statuses remain.

Important: Imputed prices and quantities are estimates, not verified transaction amounts. Negative price -5 appears on 15 remaining rows across multiple products and statuses, so the cleaned data should not be treated as fully validated accounting data. Investigate the source of these values before calculating trusted sales totals.

Original vs post-deduplication missing counts

Before removing duplicates, the original 1,000 rows had 58 missing unit prices and 41 missing quantities. After 50 exact duplicates were removed, the 950 unique rows contained 55 missing prices and 40 missing quantities. These post-deduplication values were the ones imputed.

Imputation by product

Product	Missing prices imputed	Median price used
Headphones	12	52.81
Keyboard	11	45.94
Mouse	8	47.65
Notebook	15	50.77
USB Drive	9	54.32

Deliverables

cleaned_sales.csv - primary cleaned output. **cleaning_audit.csv** - includes imputation flags. **flagged_unit_prices.csv** - rows needing price review. **cleaning_pipeline.py** - reconstructed reproducible workflow. **original_sales.csv** - synthetic source file.

This report documents a practice project based on synthetic data. The original Jupyter Notebook was not provided; the included script reconstructs the verified cleaning workflow.